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A PALEOMAGNETIC TEST OF THE PROPOSED MOJAVE–SONORA MEGASHEAR IN NORTHWESTERN MEXICO

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ABSTRACT

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Three Upper Triassic–Lower Jurassic formations in southwestern United States and northwestern Mexico have been investigated paleomagnetically as a means of testing for displacement along the proposed Mojave–Sonora megashear. Luning Formation of west-central Nevada and Sil Nakya Formation of southern Arizona lie north of the megashear whereas, in northwestern Sonora, Antimonio Formation lies south of the megashear. Previously published results from the Nazas Formation of presumed Lower Jurassic age from north-central Mexico were incorporated into the interpretations. These four units compose two sets of correlative formations on opposite sides of the proposed megashear.

Restoration of 800 km of left-lateral displacement along the megashear about the pole of rotation determined by Anderson and Schmidt (1983), improves the grouping of the two sets of paleomagnetic poles noticeably, with the improvement having statistical significance at the 95% confidence level. We feel that the data support, but do not prove, the postulated displacement along the megashear.

All four poles are displaced from Irving's (1979) smoothed apparent polar wander path for cratonic North America in the intervals 190 and 200 Ma. The displacement needed to bring all of these poles into coincidence is a pure rotation of about 8° counterclockwise about the sampling area.

Two probable Cretaceous poles were obtained from the Sand Wells Formation of southern Arizona and from unnamed volcanics in northwest Sonora. These poles are essentially identical and suggest that displacement along the megashear had ceased by the Cretaceous.

Data from the Early Jurassic Mulberry Wash and Pitoikam formations in southern Arizona appear to have been remagnetized in a later Cretaceous event and were not used in this study.

INTRODUCTION

Bullard et al. (1965) presented one of the earliest Pangea configurations by determining the best fit of the continents around the Atlantic at the 500 fathom contour line. In this classic work much of Mexico and all of Middle America were

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omitted from the reconstruction and the reason for the omission of these landmasses is evident. A sizeable overlap occurs between the Mexican–Middle American landmasses and the continent of South America if Mexico and Middle America are included in their present-day positions relative to North America. Since this early reconstruction of Pangea, geoscientists have speculated on the positions of the Mexican and Middle American landmasses relative to the rest of Pangea.

In the two decades which followed this early work, a number of Pangea reconstructions have been suggested by various workers which would accommodate the Mexican and Middle American landmasses and which would result in little or no overlap with South America (Malfait and Dinkelman, 1972; Iturralde-Vinent, 1975; Van der Voo and French, 1974; Irving, 1977; Scotese et al., 1979; White, 1980; Urrutia-Fucugauchi, 1981; Gose et al., 1982; Pindell and Dewey, 1982; Anderson and Schmidt, 1983). Paleomagnetic data which could support or negate the proposed motions are sparse at present (Urrutia-Fucugauchi, 1984). In addition, much existing data are from areas where the geology is poorly understood and local rotations can not be ruled out.

Geologic data (Anderson and Schmidt, 1983) do not support extreme rotations of Mexico relative to North America since the breakup of Pangea as some workers suggest. Therefore, those models which call for a simple, large shear of Mexico relative to North America along a major fault (Silver and Anderson, 1974; Scotese et al., 1979; Urrutia-Fucugauchi, 1981; Anderson and Schmidt, 1983) seem to be more reasonable to start with as working hypotheses.

Anderson and Schmidt (1983) propose a model which suggests a Mesozoic and Cenozoic tectonic evolution of not only Mexico, but also the landmasses which now occupy Middle America, the Gulf of Mexico, and the Caribbean Sea areas. In this model, Triassic Pangea is reconstructed with three microplates comprising the landmasses which presently lie between North America and South America. These microplates are placed in various positions to avoid the overlap which would otherwise occur in the Bullard fit. The northernmost microplate includes part of present-day northern Mexico. The northern boundary of this microplate or "Yaqui block" was a transform fault which was active in Late Jurassic time and is identified as the "Mojave–Sonora megashear" (Silver and Anderson, 1974; Anderson and Silver, 1979; Anderson and Schmidt, 1983). The southern boundary of the microplate is also postulated to have been an active transform fault which later was obscured by Cenozoic volcanism and the formation of the Trans-Mexican volcanic belt.

In their Late Triassic reconstruction of Pangea, Anderson and Schmidt (1983) place the northern Mexican block, relative to its present-day position with North America, 700 to 800 km to the northwest along the Mojave–Sonora megashear (Fig. 1b). In Late Jurassic time, 700–800 km of sinistral displacement along the Mojave–Sonora megashear resulted in the existing juxtaposition of tectonic blocks. An alternative description of this motion is to keep North America stationary and

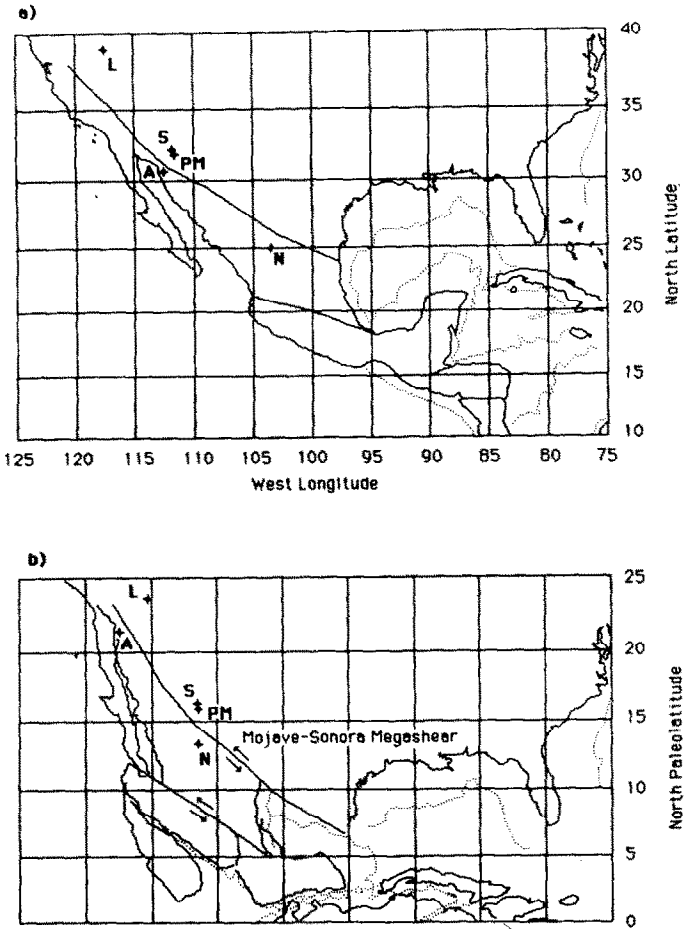


Fig. 1. a. Site map showing the locations sampled in the Luning (*L*), Sil Nakya (*S*), Pitoikam and Mulberry Wash (*PM*), and Antimonio (*A*) formations. Also shown is Nairn's (1976) site in the Nazas Formation (*N*). b. Early Mesozoic reconstruction of North America and the northern Mexico "Yaqui block" (Anderson and Schmidt, 1983). Oblique Mercator projection with map pole of projection at Irving's (1979) 200 Ma B.P. paleomagnetic pole for North America. Northern Mexico is located in its pre-displacement position, relative to North America, restored 800 km to the northwest along the Mojave–Sonora megashear.

rotate northern Mexico 15° counterclockwise relative to North America about an Euler pole of rotation at N52°, W79°.

The data in support of this model are derived from reconnaissance geologic and geochronologic studies in northwestern Mexico viewed in light of extensive research in southwestern U.S. (L.T. Silver, unpubl. data; Silver and Anderson, 1974; Anderson and Silver, 1979). The details of most of this work have yet to be published. The

most fundamental piece of geologic evidence was ascertained in the late 1960's when two juxtaposed Precambrian terranes, distinct from one another in age, were recognized in northwest Sonora, Mexico. A northwesterly-trending zone lying between the two terranes was postulated to be a relict transform fault zone. Precambrian rocks correlative to the terrane lying south of the fault zone exist in California, 700 to 800 km to the northwest of the Sonoran area. These Precambrian rocks were hypothesized to be the equivalent to the Sonoran rocks south of the fault zone, but offset along the fault. Geologic analysis of this proposed structure is continuing.

GOALS OF THIS STUDY

The primary objective of this study was to gather paleomagnetic data which would test the hypothesis of a simple, large motion along a major fault in northern Mexico in the Late Jurassic. To distinguish possible local tectonic rotations which could have occurred at each sampling site from large-scale microplate rotations, data were collected from two sets of two units on opposite sides of the Mojave–Sonora megashear that were of the same age and assumed to be correlative. In the model of Anderson and Schmidt (1983) only rocks older than late-Jurassic should record paleomagnetic poles displaced relative to equivalent age North American Paleomagnetic poles (Fig. 2). Poles determined from Cretaceous age units from northern Mexico should agree with North American Cretaceous poles if relative motion between northern Mexico and North America ceased by the Cretaceous.

This report involves analysis of data from four localities where pre-Mid-Jurassic rocks crop out (Fig. 1). The localities north of the fault are in west-central Nevada (Luning Formation) and south-central Arizona (Pitoikam, Mulberry Wash, and Sil Nakya formations). Offset equivalents to these units are proposed to exist in

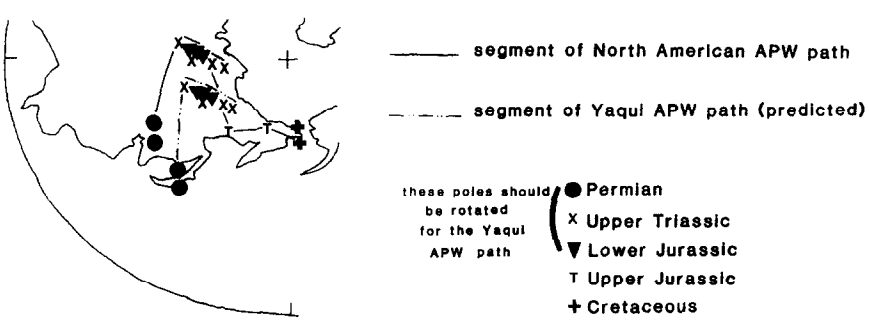


Fig. 2. Segment of North American APW path and predicted northern Mexican path for the Late Paleozoic and Mesozoic as implied by the model of Anderson and Schmidt (1983).

Sonora, northwest Mexico (Antimonio Formation) and the states of Durange, Zacatecas, and Coahuila, north-central Mexico (Nazas Formation).

At the time this study commenced, the Nazas Formation (Nairn, 1976) provided the only reliable paleomagnetic results obtained from rocks older than Mid-Jurassic and from that part of Mexico which lies south of the proposed fault zone. The Nazas Formation crops out in north-central Mexico (N25.0°, W103.5°) and consists largely of sandstone, conglomerate, shale and siltstone, with interbeds of volcaniclastic and volcanic material. Although the age of the Nazas Formation is poorly constrained, several workers (Clemmons and McLeroy, 1966; Nairn, 1976) agree on a Late Triassic–Early Jurassic age for these units.

Antimonio Formation is about 800 km to the northwest, but still south of the Mojave–Sonora megashear. This unit is composed of a thick sequence of clastics and carbonates which crop out in the northeastern part of the Sierra del Alamo in northwest Sonora (Gonzalez, 1979a, 1979b). Marine fauna from these units suggest a Late Triassic–Early Jurassic age for this miogeosynclinal sequence. Since the Antimonio and Nazas formations are separated by a geographic distance of 800 km, possible local tectonic rotations at each sampling site can be safely ruled out if their paleomagnetic poles agree.

Another approach would be to compare the poles from these two formations with poles from cratonic North America of equivalent ages. The expected deviation of the northern Mexican Early Mesozoic poles from their corresponding North American poles should be small and just barely resolvable (Fig. 2). The presence of other displaced terranes immediately to the north of the proposed megashear similar to those postulated to exist at the western margin of North America (Beck, 1976, 1980; Beck et al., 1980; Magill et al., 1981; Magill and Cox, 1981) could complicate this approach. For this reason, it was decided to make the comparisons between supposedly correlative units on opposite sides of the megashear and adjacent to it.

In order to identify formerly contiguous units to the Antimonio and the Nazas formations on the northern side of the fault zone, the reconstruction of North America and northern Mexico in Early Mesozoic time of Anderson and Schmidt (1983) was used.

On the basis of this reconstruction, which takes into account paleontologic and sedimentologic constraints, the Luning Formation, an Upper Triassic shelf deposit which crops out in the Shoshone Mountains, west-central Nevada is proposed as the North American counterpart of the Antimonio Formation (Fig. 1b). The clastic Nazas Formation probably represents a component of a magmatic arc which records subduction occurring against the craton in this region in Early Jurassic time.

The Pitoikam, Mulberry Wash, and Sil Nakya formations of the Baboquivari Mountains and Sil Nakya Hills, south-central Arizona, have recently been mapped as a thick sequence of 190 Ma. old volcanics and volcaniclastics (Haxel et al., 1978, 1980a, 1980b; Haxel, 1981; Wright et al., 1981). These units are considered to be the North American counterpart units to the Nazas Formation (Fig. 1b).

PALEOMAGNETIC TECHNIQUE

Samples were collected by drilling 2.54-cm diameter rock cores in situ employing a gasoline-powered rock drill and by collecting oriented block samples. A limited number of samples were collected during this study and our results comprise a reconnaissance. All remanent magnetization measurements were made using a 3-axis Superconducting Technology cryogenic magnetometer. All rock cores whose magnetizations were used in the final results of this study were stepwise alternating field (AF) and/or thermally demagnetized. All units sampled are tilted; therefore bed corrections were applied to all the magnetizations.

GEOLOGIC DESCRIPTIONS

Antimonio Formation of Sierra del Alamo, northwest Sonora, Mexico

Miogeosynclinal rocks of Permian and Triassic–Jurassic age crop out in northwest Sonora and are well exposed in the northeastern part of Sierra del Alamo

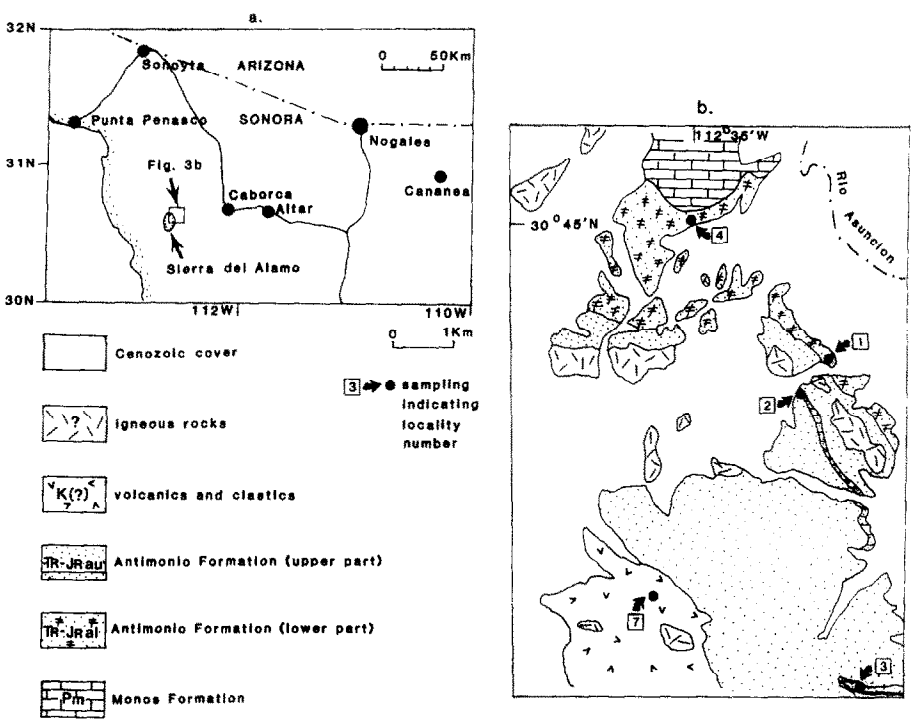


Fig. 3. Location and generalized geologic maps of sampled area in Sierra del Alamo, northwest Sonora, Mexico. Geology generalized from Gonzalez (1979b).

about 30 km west of the town of Caborca (Fig. 3). Beds of limestone and shaly limestone comprise the lowest 500-m-thick unit called the Monos Formation (Gonzalez, 1979a, 1979b). Marine fauna from these units indicate a Mid-Permian, Leonardian age for this formation (Cooper and Arellano, 1946; Gonzalez, 1979a, 1979b). Overlying these predominantly limestone beds is a 3400-m-thick sequence of sandstone, siltstone, mudstone, and limestone. Marine fauna collected from these beds yield a Late Triassic–Early Jurassic age. This sequence mapped locally as the Antimonio Formation is considered to be equivalent to the Barranca Group of eastern Sonora (Alencaster de Cserna, 1961).

The Permian and Triassic–Jurassic strata are disconformably overlain by a volcanic and volcaniclastic sequence of presumed Cretaceous (?) age based on regional stratigraphic constraints. Numerous intrusions whose exact ages are unknown cut the sequence.

The Permian, Triassic–Jurassic, and Cretaceous (?) beds generally strike northwesterly and dip 10° to 60° to the southwest.

Sampling was concentrated in beds of the Antimonio Formation. On the basis of marine fauna contained within these beds (Gonzalez, 1979a, 1979b; Fig. 3), the formation has been divided into the Lower Antimonio Formation (Upper Triassic; 1300 m) and the Upper Antimonio Formation (Lower Jurassic; 2100 m). Units from four different stratigraphic levels in the formation were sampled. Samples collected from beds of red siltstone (locality 4) and limestone (locality 1) are Upper Triasssic. These localities are in the basal beds and upper beds from the lower part of the formation (Fig. 3). Samples collected from predominantly arenaceous limestone and sandstone beds (locality 2) and gray mudstone (locality 3) are early Jurassic in age and represent the lower-upper part and middle-upper portion of the formation, respectively (Fig. 3).

Less extensive sampling was done in beds within the unnamed Cretaceous (?) volcanic sequence (locality 7) overlying the Antimonio Formation.

Luning Formation of Shoshone Mountains, west-central Nevada

Beds of the Upper Triassic Luning Formation crop out in west-central Nevada. Much of the Luning Formation is considered allochthonous (Muller and Ferguson, 1939; Ferguson and Muller, 1949; Speed, 1978; Reilly et al., 1980). However, according to the work of Silberling (1959), Luning Formation is autochthonous in the Shoshone Mountains (Fig. 4).

In the Shoshone Mountains, the stratigraphy is diverse. In ascending stratigraphic order, the Pablo Formation consists of clastic, limestone, and greenstone beds (> 800 m) which are assigned a very tentative Permian (?) age. Disconformably overlying these units, the Grantsville Formation consists of clastic and limestone beds (250 m) tentatively assigned mid-Triassic (?) age. The Luning Formation disconformably overlies the Grantsville Formation and consists of a lower clastic

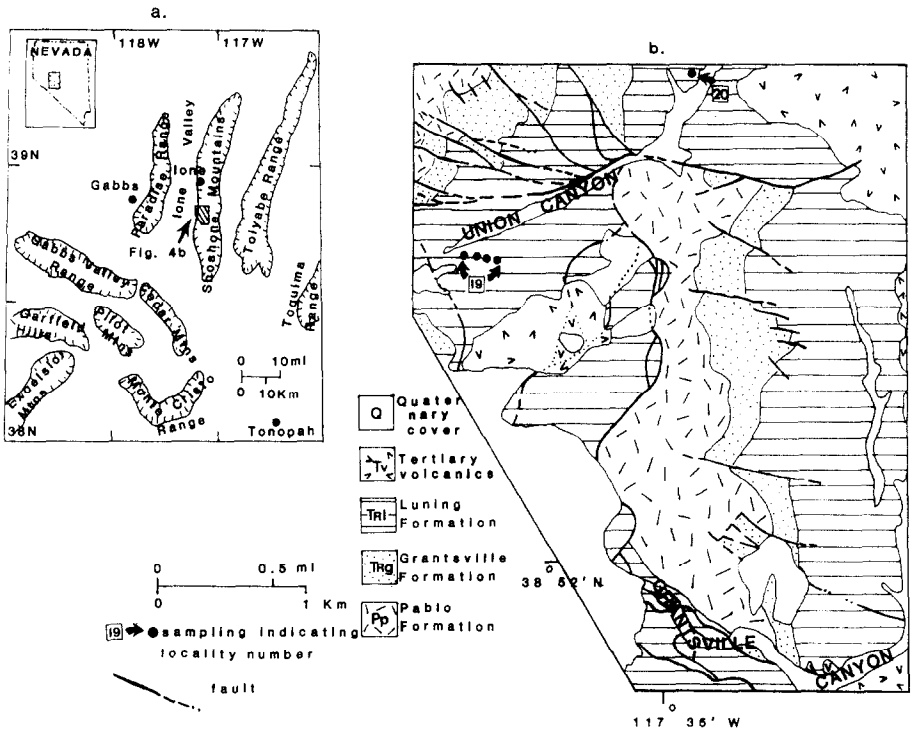


Fig. 4. Location and generalized geologic maps of samples area in Shoshone Mountains, west-central Nevada. Geology generalized from Silberling (1959).

member (200 m), a shaly limestone member (200 m), a calcareous shale member (180 m), and an upper carbonate member (670 m). The age of the Luning Formation is clearly established as Upper Triassic because of abundant marine fossils in the beds. The undifferentiated Sunrise and Gabbs formations consisting predominantly of siltstone and limestone beds (900 m) conformably overlie the Luning Formation. Meager fossil data from these formations suggest they may be assigned to the Lower Jurassic.

The Mesozoic sequence is completed by limestone and clastic beds which comprise the Middle Jurassic age (?) Dunlap Formation that conformably overlies the Sunrise and Gabbs formations. The total thickness of the Sunrise, Gabbs, and Dunlap formations, which crop out in an area south of that shown in Fig. 5, probably exceeds 750 m. Tertiary volcanic rocks overlie these Paleozoic and Mesozoic units, and intrusive rocks of probable Mesozoic and Tertiary age crop out in the area.

In Union Canyon (localities 19 and 20; Fig. 4), Paleozoic and Mesozoic rocks generally strike northerly with low to moderate dips to the east. Silberling (1959) claims the rocks in this area of the Shoshone Mountains comprise part of the

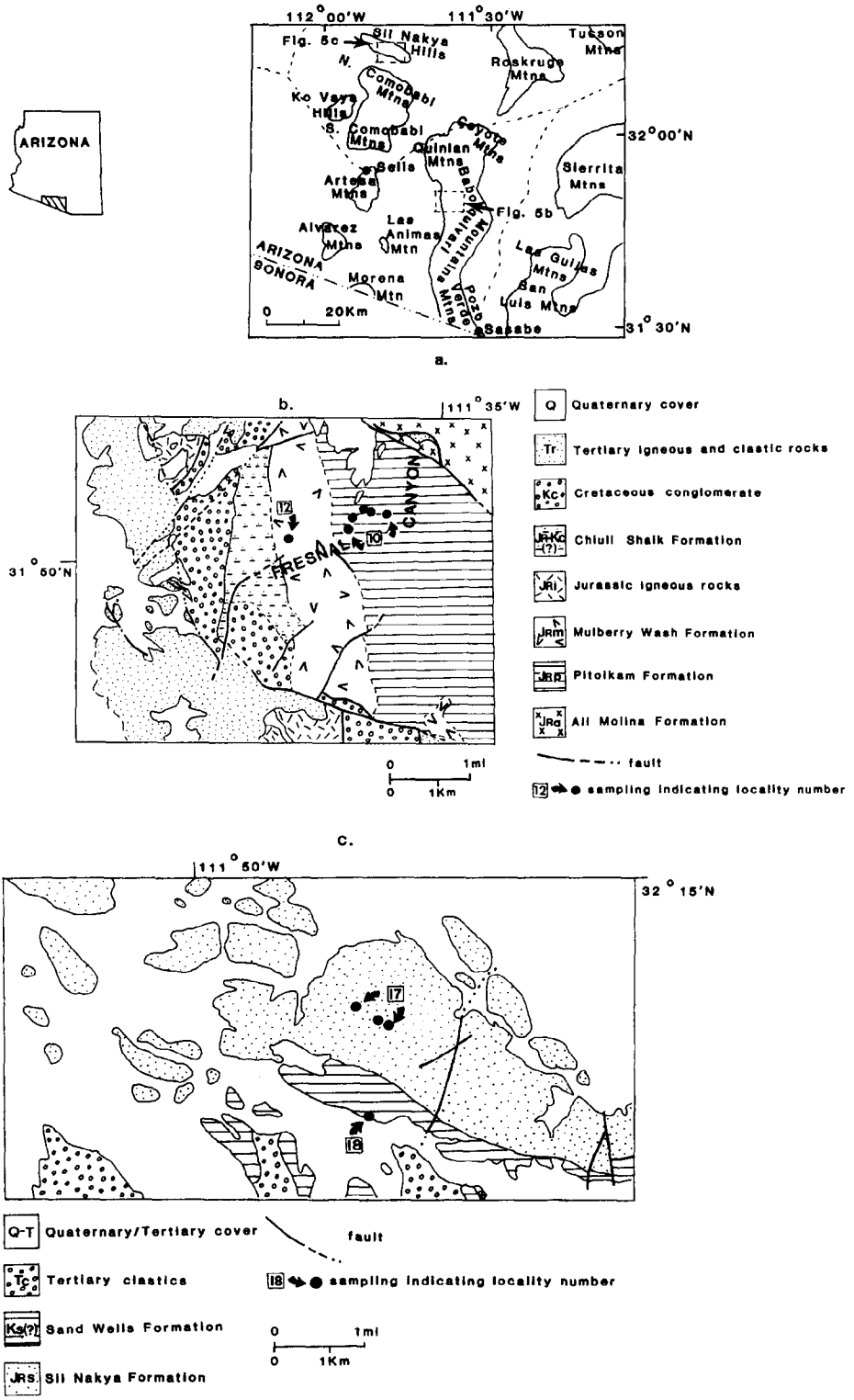


Fig. 5. Location and generalized geologic maps of sampled areas in south-central Arizona. Geology generalized from Haxel et al. (1978, 1980a).

upright limb of an overturned, north-trending anticline. Samples were collected exclusively in the calcareous shale and shaly limestone members of the Upper Triassic Luning Formation (localities 19 and 20; Fig. 4).

Pitoikam, Mulberry Wash, and Sil Nakya formations of the Baboquivari Mountains and Sil Nakya Hills, south-central Arizona

Recent reconnaissance geologic mapping and uranium–lead isotopic studies in south-central Arizona have provided valuable insight into the geology of the Baboquivari Mountains and Sil Nakya Hills (Haxel et al., 1978, 1980a, 1980b; Haxel, 1981; Wright et al., 1981). In the Baboquivari Mountains, which are partially comprised of Mesozoic supracrustal rocks (Fig. 5), a thick sequence of volcanic and volcaniclastic rocks dated by the uranium–lead isotopic method yields ages of 190 Ma. from zircons (Wright et al., 1981). Almost 30 km north-northwest of the Baboquivari Mountains, similar volcanics and volcaniclastics, also dated at 190 Ma. (Wright et al., 1981), crop out in the Sil Nakya Hills. On the basis of similar lithology and age, the Early Jurassic rocks in the Sil Nakya Hills are thought to be the equivalent rocks to the Early Jurassic volcanics exposed in the Baboquivari Mountains. These volcanic accumulations represent part of a magmatic belt that existed in south-central Arizona during the Early Jurassic.

In the Baboquivari Mountains, the rocks can be divided into four broad units (Haxel et al., 1980b). In ascending stratigraphic order these units consist of: (1) Lower Mesozoic supracrustal rocks; (2) Middle and Late Jurassic plutonic rocks; (3) Upper Mesozoic supracrustal rocks which rest upon the Jurassic plutons, and (4) Late Cretaceous and Early Tertiary granitic rocks (Fig. 5; Haxel et al., 1980b). Late Cretaceous regional metamorphism and/or mid-Tertiary metamorphism have heated these rocks in many areas. Within the Mesozoic supracrustal rocks the stratigraphic sequence consists of three Early Jurassic volcanic and volcaniclastic formations (Ali Molina, Pitoikam, and Mulberry Wash formations; combined thickness > 7000 m) unconformably overlain by the Jurassic–Cretaceous (?) Chiuli Shaik Formation, also mainly volcanics and clastics. These units are unconformably succeeded by Cretaceous–Tertiary (?) conglomerates and Tertiary sedimentary and volcanic rocks. Jurassic, Cretaceous, and Tertiary granites cut the sequence (Fig. 5b).

The units in the Baboquivari Mountains generally strike to the north with rather steep dips to the west forming a homoclinal structure. Low and high-angle faults cut the sequence.

Sampling was concentrated in the Fresnal Canyon area of the Baboquivari Mountains in the Early Jurassic Pitoikam Formation and overlying Mulberry Wash Formation (localities 10 and 12; Fig. 5b). The Ali Molina Formation records regional metamorphism and therefore was not sampled.

The Sil Nakya Hills are almost 30 km north-northwest of the Fresnal Canyon area (Fig. 5c). Here the stratigraphy is less diverse than in the Baboquivari

Mountains and consists of the Early Jurassic Sil Nakya Formation composed mainly of volcanics and volcaniclastics (2500 m), unconformably overlain by the Cretaceous (?) Sand Wells Formation, also composed of volcanics and clastics (1900 m). The Sand Wells Formation is succeeded by an unnamed Cretaceous (?) ash-flow tuff unit.

These units in the Sil Nakya Hills strike northwesterly with moderate dips to the southwest. High-angle faults cut the sequence.

Sampling in the Sil Nakya Hills was concentrated in the Early Jurassic Sil Nakya Formation (locality 17; Fig. 5c). Less extensive sampling was also done in the Cretaceous (?) Sand Wells Formation (locality 18; Fig. 5c).

PALEOMAGNETIC RESULTS

The paleomagnetic results from the three areas sampled in this study are condensed into Table 1. AF and thermal demagnetization techniques, IRM acquisition studies, and polished thin-section studies were used to identify the carriers of magnetic remanence. Because the lithologies investigated are highly diverse, information regarding the carriers of remanence for these rocks has been condensed into Table 1A.

Rocks commonly record more than one magnetic component (direction). If the magnetization of a rock sample is multi-component, it may be difficult to identify which component represents the field direction present at the time the rock formed. Invariably, a partly subjective judgement must be made by comparing all the magnetic results, taking the age of the rocks into consideration, and looking closely at the geology of the area sampled.

Zijderveld (1967) diagrams are an invaluable tool in distinguishing single-component samples from multi-component samples, and in interpreting the components observed. The multi-component or single-component nature of the units studied is information which has been added to Table 1A. This table describes whether the majority of samples used in the determination of the mean paleomagnetic pole for any one unit are multi-component (containing more than one field direction, excluding viscous low-coercivity components).

Types of magnetic behavior

Behavior A

Samples which display magnetic behavior A have Zijderveld diagrams (see Fig. 6a) which show essentially single-component behavior when AF demagnetized. Excluding a possible small viscous component which is usually removed by 15 mT peak field, the horizontal and vertical components of the field direction are seen to head toward the origin of the diagram with only minor deflections in direction. This single component generally reaches or almost reaches the origin of the diagram by

TABLE 1

Paleomagnetic data

A. Magnetic characteristics

Unit name and age	Primary magnetic carriers of characteristic direction		Samples multi component? (excluding low coercivity viscous component)		Example Zijderveld diagram which best displays magnetic behavior (A, B, C or D)	Mean field direction (I_m, D_m) for unit determined by: Af demagnetization (H: mT = peak field) and/or thermal demagnetization (T = °C)	
	medium coercivity 5–140 mT	high coercivity >140 mT	yes	no		minimum dispersion criteria	line fit to medium coercivity on Zijderveld diagram
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
1. Sierra del Alamo, northwest Sonora, Mexico							
Unnamed volcanics and volcaniclastics, locality 7	×	×		×	A, C, and D	H = 20, 40, and 500 mT	
Antimonio Fm: gray mudstones, locality 3	×		×		A		
Arenaceous limestone and limey sandstone, locality 2	×	×	×		B and C		
Limestone, locality 1	×		×		B		×
Red siltstones, locality 4		×		×	C	T = 350°C	

2. Shoshone Mountains, west-central Nevada							
Luning Fm: shaly limestones locality 19	×		×	×			×
Shaly limestone locality 20	×		×				×
3. Baboquivari Mountains and Sil Nakya Hills, south-central Arizona							
Mulberry Wash Fm: volcanics and volcanic mudstone, locality 12	×	×		×	C and D	H—20 through 140 mT	
Pitoikam Fm: volcaniclastic epiclastic sandstone and siltstone, locality 10	×	×		×	C and D	H—40 mT	
Sand Wells Fm: maroon sandstones and siltstones, locality 18	×			×	A	H—30 mT	
Sil Nakya Fm: volcanics locality 17	×	×		×	A, C, and D	T—650°C	×

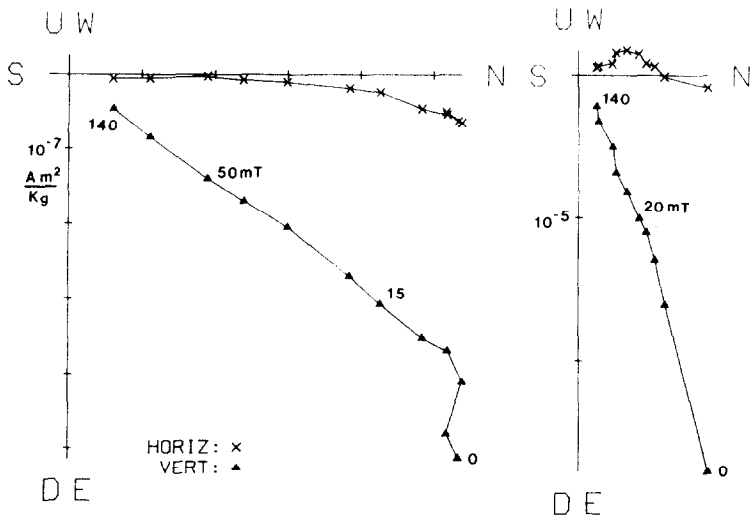
B. Paleomagnetic results *

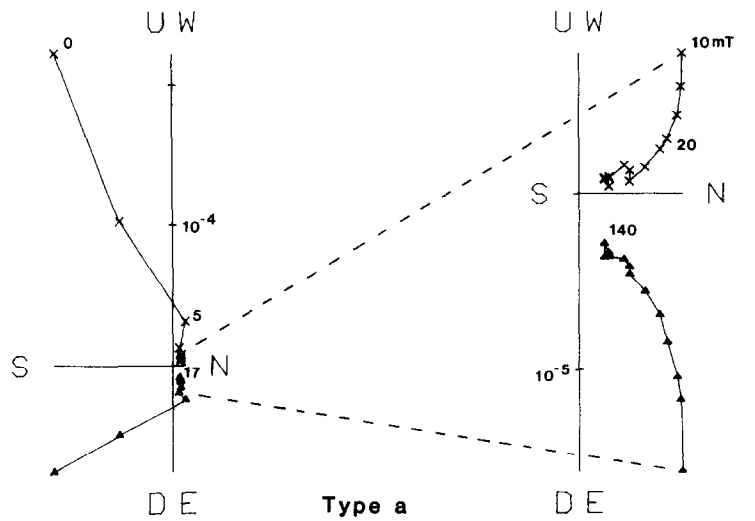
Unit name and age	Age of rocks according to **			λ_s,	ϕ_s	I_m,	D_m	λ_m,	ϕ_m	α_{95}	K	d_p,	d_m	R	N/n	% N represents of total number of samples collected from locality	Apparent age of magnetization
	fossils	radiometric dating	stratigraphic relation														
(1)	(2)	(3)	(4)	(5)		(6)		(7)		(8)	(9)	(10)		(11)	(12)	(13)	(14)
1. Sierra del Alamo, northwest Sonora, Mexico																	
Unnamed volcanics and volcaniclastics, locality 7			Kr(?)	N30.67,	W112.61	65.6,	327.3	N59.8,	W158.7	9.4	97.2	7.5,	12.4	3.97	4/62	66	Kr
Antimonio Fm: gray mudstones, locality 3	E. Jr.			N30.65,	W112.54	32.5,	0.1			7.2	87.2			5.94	6/26	50	E. Jr
Arenaceous limestone and limey sandstone, locality 2	E. Jr.			N30.72,	W112.54	−31.2,	165.4			23.6	11.5			4.65	5/8	38	E. Jr.
Limestone, locality 1	U. Tr			N30.72,	W112.54	35.5,	302.2			8.3	26.1			12.50	13/24	43	U. Tr
Red siltstones, locality 4			U. Tr.	N30.75,	W112.58	−37.0,	164.4			13.8	17.0			7.59	8/81	100	U. Tr
Pole I ($B = 4$) (units at localities 3, 2, 1 and 4)	U. Tr.	E. Jr.		N30.72,	W112.54	36.0,	338.7	W68.0,	E133.1	23.8	15.8	16.1,	24.0	3.81	32/139	58	U. Tr.–E. Jr
Pole II ($B = 3$) (units at localities 3, 2 and 4)	U. Tr.	E. Jr.		N30.72,	W112.54	33.8,	350.0	N74.8,	E106.3	12.1	104.4	7.9,	11.7	2.98	19/115	47	U. Tr.–E. Jr.

2. Shoshone Mountains, west-central Nevada															
Luning Fm: Shaly limestones, locality 19	U. Tr.			N38.87, W117.59	36.4,	2.8			3.5	60.9			28.5	29/41	100	U. Tr.
Shaly limestone, locality 20	U. Tr.			N38.88, W117.58	41.5,	2.5			5.8	90.7			7.92	8/19	100	U. Tr.
Pole ($B = 2$)	U. Tr			N38.87, W117.59	39.0,	2.7	N73.0,	E54.0	11.2	503.9	7.9,	10.4	2.0	37/60	100	U. Tr.
3. Sierra del Alamo, northwest Sonora, Mexico															
Mulberry Wash Fm: volcanics and volcanic mudstone, locality 12		E. Jr.		N31.84, W111.62												
Pitoikam Fm: volcaniclastic epiclastic sandstone and siltstone, locality 10		E. Jr.		N31.84, W111.60												
Combined-uncorrected		E. Jr.		N31.8, W111.6	69.4,	334.0	N61.5,	W114.9	6.1	40.3	4.0,	8.9	14.7	15/22	94	Kr
Combined-corrected		E. Jr.		N31.8, W111.6	25.7,	261.3	S 0.1,	E174.4	11.2	21.7	6.5,	72.1	13.9	15/22	94	?
Sand Wells Fm. maroon sandstones and siltstones locality 18			Kr(?)	N32.21, W111.80	69.3,	321.3	N55.4,	W153.3	8.7	48.9	6.7,	12.7	6.88	7/21	100	Kr
Sil Nakya Fm: volcanics, locality 17		E. Jr.		N32.23, W111.80	31.3,	356.7	N74.4,	E80.1	7.6	37.0	4.8,	7.3	10.7	11/33	100	E. Jr.

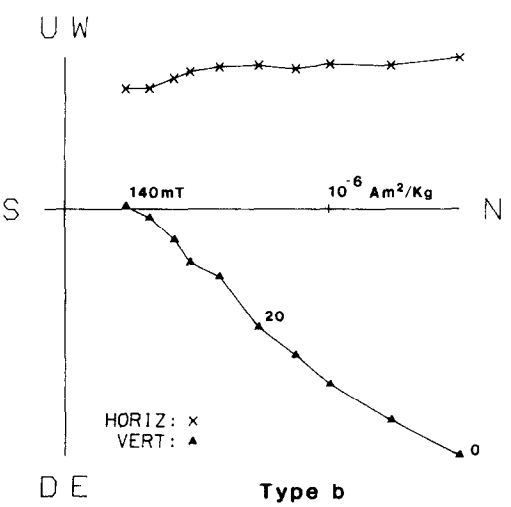
* Explanation of symbols: λ_s and ϕ_s = latitude and longitude of sampling locality, I_m and D_m = mean inclination and declination, λ_m and ϕ_m = latitude and longitude of paleomagnetic pole, α_{95} = radius of circle of confidence about field direction at 95% level, K = precision parameter, d_p and d_m = semi-major and minor axes of oval of confidence about paleomagnetic pole at 95% level, R = resultant vector, N/n = number of samples used in Fisher analysis/number of specimens cut from N samples. Sample mean field directions were weighted equally to determine unit mean field direction.

** Based on remanence corrected for dip of beds unless otherwise specified.


U W
S
N
140
10^{-7}
$\frac{Am^2}{Kg}$
50mT
15
0
HORIZ: ×
VERT: ▲
D E
U W
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140
10^{-5}
20 mT
0
D E


U W
0
10^{-4}
5
S
17
N
D E
U W
10mT
20
S
N
140
10^{-5}
D E

Type a


U W
140mT
10^{-6} Am²/Kg
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Type b

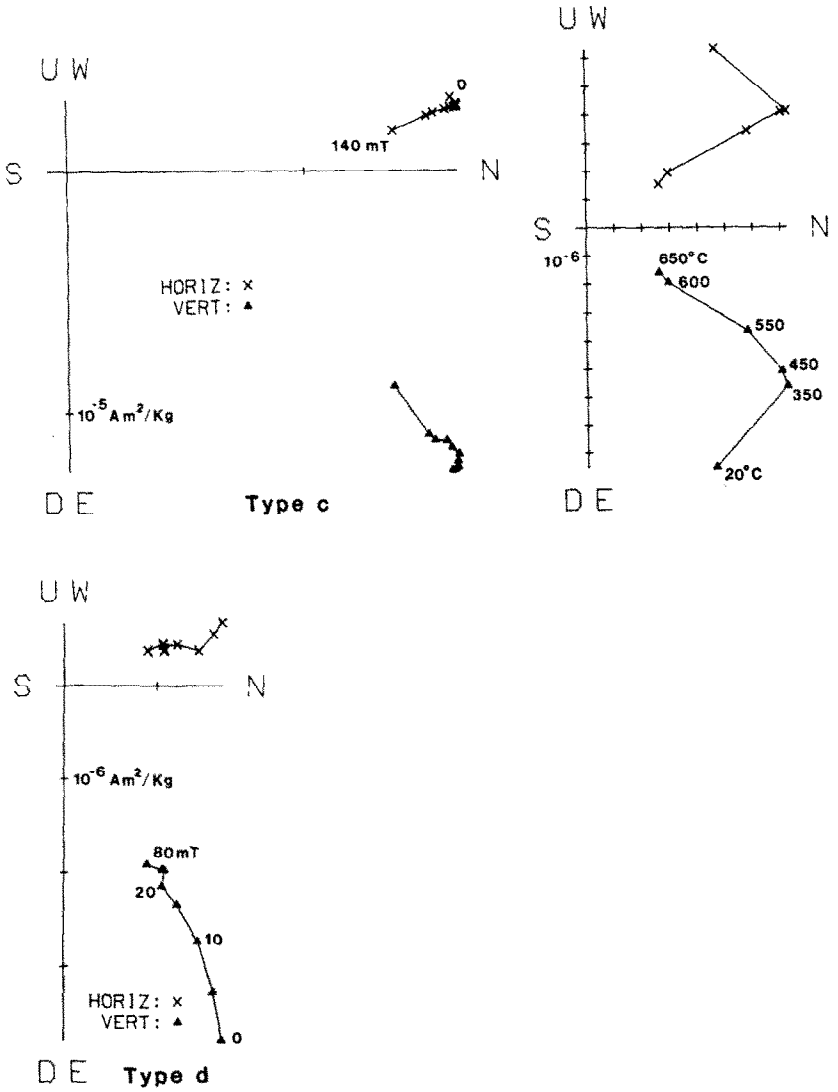


Fig. 6. Zijderveld diagrams of specimens which display characteristic behaviors A, B, C, and D. Af demagnetization peak field values in mT. Thermal demagnetization in °C. a. Magnetic behavior A. b. magnetic behavior B. c. magnetic behavior C. d. magnetic behavior D.

140 mT peak field. The magnetization of these samples is therefore dominated by the medium coercivity-range magnetic minerals, probably titanomagnetites, which have recorded essentially a single component.

Behavior B

Behavior B is similar to behavior A, except that behavior B samples are multi-component, with the additional component contributed in the high coercivity range. The field direction contributed by this high coercivity component is not the

same direction as that recorded by the medium coercivity and is probably due to secondary hematite. For the samples that display behavior B, a three-dimensional line not constrained to pass through the origin was fitted to the direction recorded by the medium coercivity magnetic minerals. This direction was evaluated and considered the original magnetization recorded by the rocks as they formed.

Behavior C

Samples which display behavior C are dominated by a high coercivity component when AF demagnetized. During the AF demagnetization sequence up to peak fields of 140 mT, little decrease in intensity of the magnetization occurs. When thermal demagnetization studies are also conducted, behavior C samples display primarily one component. Hematite is the most likely candidate as the magnetic mineral responsible for the remanent magnetization of these samples.

Behavior D

When subjected to AF demagnetization, samples which display behavior type D appear to contain both medium and high coercivity components which contribute approximately equal amounts to the magnetization. The medium coercivity range forms a line headed for the origin of the Zijderveld diagram. Because of the nature of deformation these rocks experienced, they often contained fine fractures and did not remain intact through stepwise thermal demagnetization. These samples could not be demagnetized to negligible intensities to observe their behavior on Zijderveld diagrams. The behavior of these samples through AF demagnetization is suggestive that these samples are mostly single-component with probable magnetic carriers of titanomagnetite and hematite.

With the variety of magnetic behaviors observed, it is difficult to identify the field directions which represent the ambient field when the rocks formed. For a given rock unit sampled, after examining the demagnetization behavior on Zijderveld diagrams for samples studied, along with using Fisherian statistical analysis of the samples collected from the rock unit, judgement was made as to whether the rock unit preserved a consistent field direction as observed in the majority of samples measured. For all of the rock units investigated in this study, both the tilt corrected and uncorrected field directions were determined and evaluated. Criteria used to discern magnetizations acquired during or shortly after the formation of the rocks from those field directions representing remagnetizations include the following: improvement in the clustering of directions with tilt correction, the presence of antipodal field directions representing mixed polarities within a formation, and proximity of the determined paleomagnetic poles to expected pole positions for cratonic North America of the appropriate age.

The mean field direction for a unit was determined by one of two methods. In the first method of data treatment, the statistical approach of finding the demagnetization level at which the sample field directions show minimum dispersion was

utilized. The sample field directions at this demagnetization level were used in a Fisher statistical analysis to determine a mean inclination and declination for the unit. This method was used for single-component samples that displayed behavior types A, C, and D.

The second method of data treatment was used when a well-defined line segment was observed on the Zijderveld diagram for most of the samples from the unit in question. A line was fitted to the horizontal and vertical components of the Zijderveld diagram for a specified demagnetization interval (always medium coercivity range). A fitted inclination and declination which best represents the field direction recorded by the rock core was obtained from this line. This method of data analysis was used for samples which displayed types A and B (Fig. 6). The fitted field directions were used in a Fisher statistical analysis to determine the mean field direction for the unit.

If samples did not display a characteristic behavior (A, B, C, or D) or if the recorded field direction appeared to represent a remagnetization of the rocks, the samples were rejected in the final discussion of the data.

In Table 1A the information in the last two columns describes whether the mean field direction obtained for a unit was determined by the minimum dispersion criterion or by using fitted data. The information in this column also describes whether samples were primarily AF or thermally demagnetized. The sample directions and formation mean directions for the unnamed volcanics from Sonora and for the Sand Wells, Antimonio, Sil Nakya, and Luning formations are shown plotted on equal area projections in Figs. 7 and 8.

Paleomagnetic poles

The mean field direction (I_m, D_m) determined for a unit within a formation was used to calculate the paleomagnetic pole (λ_m, ϕ_m) for the formation. For Antimonio and Luning formations multiple stratigraphic units were sampled. In these cases, each sampled unit was weighted equally in the Fisher statistical analysis to determine the pole for the formation. For the Antimonio Formation the sample directions from four units (Fig. 7) yield a non-Fisherian distribution (Table 1, Pole I). Local structural complexities at locality 1 not recognized in the field may account for these results. The pole was recalculated (Table 1, Pole II), excluding the mean direction of the discrepant unit. Only the latter pole is used in the final discussion of the results.

The results of limited samplings of unnamed volcanics and Sand Wells Formation are given in Table 1 and in Fig. 8. Results from the Pitoikam and Mulberry Wash formations in Table 1 have been combined because both formations are 190 Ma old (Wright et al., 1981) and comprise a continuous stratigraphic section.

In Table 1B the information in columns (1–4) and (14) describes the best determined age of the formation and the apparent age of the magnetization,

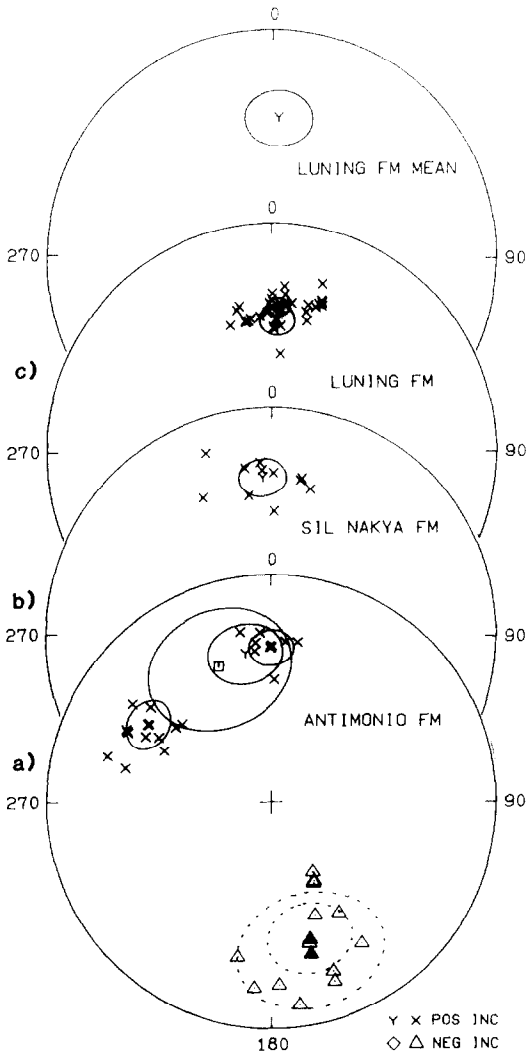


Fig. 7. Equal area projections of Antimonio, Sil Nakya, and Luning formation field directions and means with ovals of confidence. X—positive inclination; triangle—negative inclination. a. Antimonio Formation field directions. Four stratigraphic intervals sampled. Mean direction of each interval shown by bolder X or solid triangle. Mean of four intervals shown by open square and mean of three intervals shown with Y symbol. b. Sil Nakya Formation field directions. Mean shown with Y symbol. c. Luning Formation field direction. Two stratigraphic intervals sampled. Mean of each shown with bolder X; mean of two means shown with Y symbol.

respectively. For the combined results of the Pitoikam and Mulberry Wash formations these ages do not coincide; therefore remagnetization of these formations is implied, assuming that major rotations have not taken place. The results suggest that remagnetization took place in Cretaceous time, after tilting of the units.

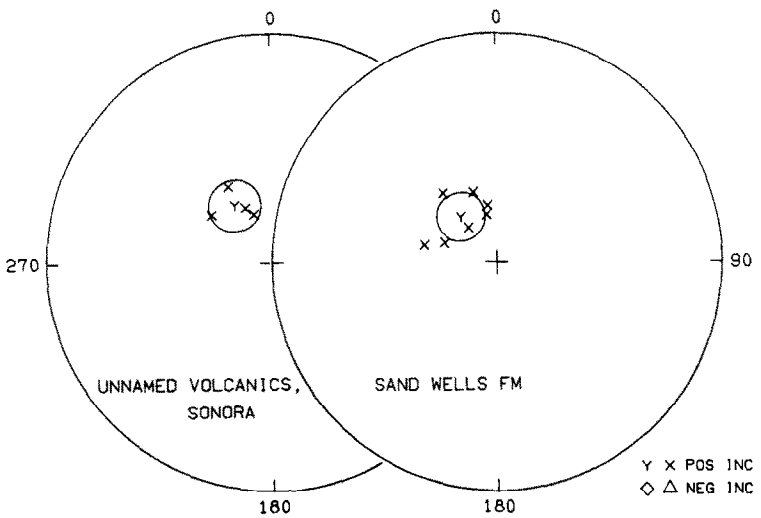


Fig. 8. Equal area projection of unnamed volcanics from Sonora and Sand Wells Formation field directions and means with ovals of confidence. X = positive inclination. Y = mean.

Remagnetization may be related to extensive Late Cretaceous–Early Tertiary regional metamorphism associated with magmatic events in the area. Results from the Pitoikam and Mulberry Wash formations were discarded for this reason.

Significance of Cretaceous paleomagnetic poles

Discussions of displaced terranes in the North American Cordillera have been prevalent in the literature. Displacement of some of these terranes took place in the Cenozoic (Beck, 1976, 1980; Howell, 1980; Beck et al., 1981). Paleomagnetic data from Cretaceous rocks are necessary to distinguish large post-Cretaceous displacements from earlier translations. If Cretaceous rocks yield cratonal poles, then displacements recorded in order rocks may safely be assumed to be pre-Cretaceous.

Paleomagnetic data (Fig. 9) for Cretaceous age rocks from three of the four sample localities have been collected (Fig. 1). Additional data from south-central Arizona (Roskruge volcanics; Vugteveen et al., 1981) and north-central Mexico (Difunta Group; Nairn, 1976) exist in previously published work. The Shoshone Mountains, west-central Nevada, where the Luning Formation was sampled, is the only area from which no previous paleomagnetic data for Cretaceous rocks are available.

The Cretaceous paleomagnetic poles determined for rocks which crop out in northwest Sonora, Mexico, north-central Mexico, and south-central Arizona overlap North American Cretaceous poles at the 95% confidence level (Fig. 9). There is considerable scatter of these poles about the mean North American Cretaceous pole, but the agreement between the poles for the Sand Wells Formation in south-central

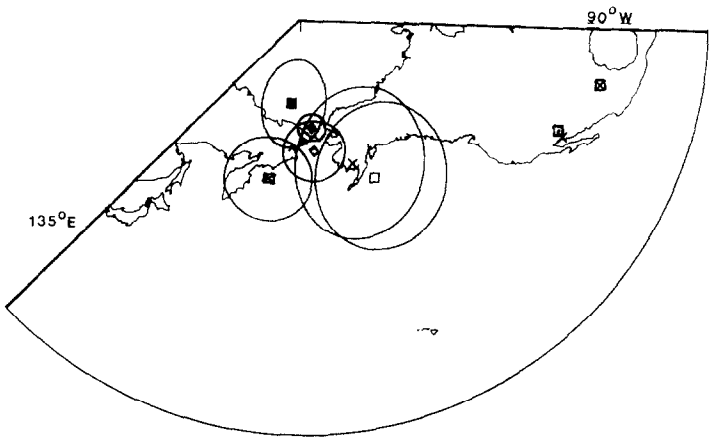


Fig. 9. Cretaceous pole positions from units located in area near the Antimonio, Nazas, and Sil Nakya formations. Paleomagnetic poles are shown with circles or ovals of confidence and sampling sites shown with same symbol as the pole. Open square = Sand Wells Formation, south-central Arizona (this study). Solid square = Roskruge volcanics, south-central Arizona (Vugteveen et al., 1981). X—Unnamed volcanics, northwest Sonora (this study); boxed X—Difunta Group, north-central Mexico (Nairn, 1976); open diamond—North American pole (McElhinny, 1973); closed diamond—mean of eight North American poles (Irving, 1979).

Arizona and the unnamed volcanics from northwest Sonora (locality 7), both from this study, is noteworthy because these sites lie on opposite sides of the megashear. If relative displacements occurred since the Cretaceous, they are not on the order of thousands of kilometers as postulated to have occurred in some areas (e.g. Beck et al., 1981). However, because of the reconnaissance nature of most of this Cretaceous data, displacemens or rotations of these areas of a small order are possible. Large-scale post-Cretaceous displacements of northwest Sonora, north-central Mexico, and south-central Arizona relative to cratonic North America and to each other are not indicated.

Significance of Late Triassic–Early Jurassic paleomagnetic poles

Because large post-Cretaceous displacements of the sampled areas are not indicated, Early Mesozoic poles obtained for northern Mexico and North America can be compared with greater confidence. Figure 10a shows the location of the four observed Early Mesozoic paleomagnetic poles of interest to this study (Table 2) along with their associated ovals of confidence at the 95% level. The mean early Mesozoic pole for northern Mexico determined from the Antimonio and Nazas formations lies at N75.7°, E112.4° ($N = 2$, $\alpha_{95} = 7.8$, $k = 1027$). The mean early Mesozoic pole for North America determined from the Luning and Sil Nakya formations, which are correlative with the northern Mexican formations, lies at N74.1° E 66.5° ($N = 2$, $\alpha_{95} = 16.2$, $k = 240$).

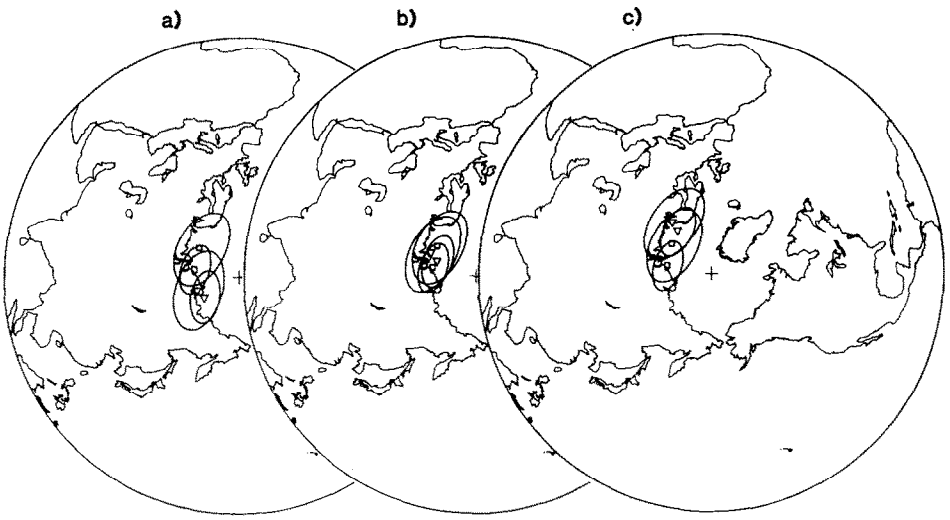


Fig. 10. Comparison of observed and modeled paleomagnetic poles and ovals of confidence determined for Early Mesozoic equivalent units from northern Mexico and the U.S. Western Cordillera. Triangles—northern Mexican Antimonio and Nazas formation poles; open squares—U.S. Western Cordilleran Luning and Sil Nakya formation poles. a. Observed paleomagnetic poles. b. Observed U.S. Western Cordilleran poles, but northern Mexican poles have been rotated about pole of rotation suggested by tectonic model of Anderson and Schmidt (1983). c. Observed U.S. Western Cordilleran poles, but northern Mexican poles have been rotated about pole of rotation suggested by tectonic model of Scotese et al. (1979).

All ovals and circles of confidence for poles plotted in Fig. 10a overlap one another at the 95% confidence level. Therefore, these paleomagnetic poles are statistically indistinguishable. Nonetheless, the mean northern Mexican early Mesozoic pole is slightly displaced a few degrees counterclockwise relative to the North American poles. This displacement has the proper magnitude and sense of displacement of the proposed Mojave–Sonora megashear. Tectonic models suggested by Scotese et al. (1979) and Anderson and Schmidt (1983) were chosen for comparison with the data because they are of a quantitative nature and provide an Euler pole and angle of rotation between North America and northern Mexico (Table 2; Fig. 10b, c).

First, the displacements between North America and northern Mexico called for in each model were restored. By keeping North America stationary, northern Mexico was restored to its pre-breakup position by rotating the northern Mexico block about the Euler pole of rotation by the amount called for in each model (Table 2). Likewise, the observed northern Mexican paleomagnetic poles were rotated in the same amount and sense (Fig. 10a, b). The models of Scotese et al. (1979) and Anderson and Schmidt (1983) use similar Euler poles of rotation (Table 2). Additionally, both models call for a counterclockwise rotation of Mexico relative

TABLE 2

Observed and modeled paleomagnetic data for Early Mesozoic units of interest to this study

Parameters [a]	Northern Mexico		U.S. Western Cordilleran	
	Antimonio (pole II)	Nazas Fm.	Luning Fm.	Sil Nakya Fm.
Observed				
Site coordinates				
λ_s	30.72	25.5	38.87	32.23
ϕ_s	−112.54	−102.9	−117.59	−111.80
I_m	33.8	28.4	39.0	31.3
D_m	350.0	350.6	2.7	356.7
α_{95}	12.1	8.2	11.2	7.6
Pole				
λ_m	74.8	76.4	73.0	74.4
ϕ_m	106.3	119.2	54.0	80.1
Modeled				
(a) to restore displacement of northern Mexico suggested by Anderson and Schmidt (1983) [b]				
Site coordinates				
λ'_s	36.3	29.8	–	–
ϕ'_s	−119.5	−110.5	–	–
I'_m	33.8	28.4	–	–
D'_m	358.8	358.8	–	–
Pole				
λ'_m	72.2	75.3	–	–
ϕ'_m	64.1	70.2	–	–
(b) to restore displacement of northern Mexico suggested by Scotese et al. (1979) [c]				
Site coordinates				
λ''_s	39.7	32.1	–	–
ϕ''_s	−121.8	−113.8	–	–
I''_m	33.8	28.4	–	–
D''_m	7.9	8.6	–	–
Pole				
λ''_m	67.8	71.3	–	–
ϕ''_m	38.1	39.3	–	–

[a] Symbols are the same as those used for Table 1.

[b] Model of Anderson and Schmidt (1983) suggests the Euler pole of rotation between northern Mexico and North America is N52°, W79°. Angle needed to restore rotation = $\omega = -15°$. $-\omega$ = clockwise rotation.

[c] model of Scotese et al. (1979) suggests Euler pole of rotation between northern Mexico and North America is N48.6°, W85.9°. Angle needed to restore rotation = $\omega = 26°$

to North America to restore Mexico to its pre-breakup position. The primary difference is that the model of Scotese et al. (1979) calls for a larger angle of rotation (−26°) which is primarily based on geometrical constraints, whereas the

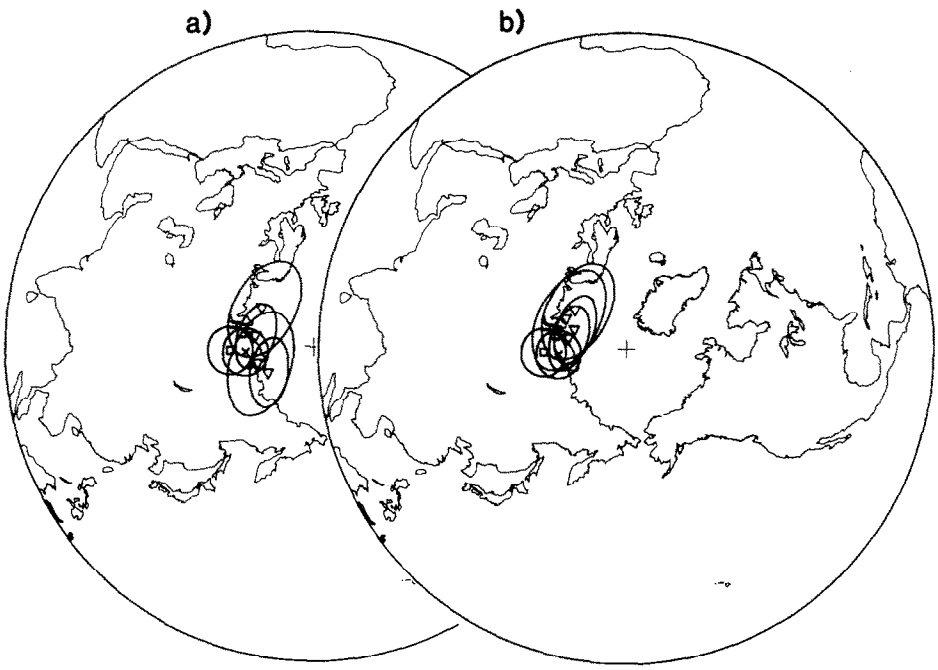


Fig. 11. Comparison of observed and modeled Antimonio, Luning, Nazas, and Sil Nakya formation paleomagnetic poles to North American cratonic 200 and 190 Ma paleomagnetic poles. Triangles—poles for above named units; square—200 Ma B.P. pole and X—190 Ma B.P. pole for North America (Irving, 1979). a. Observed paleomagnetic poles. b. Observed Luning and Sil Nakya poles. Antimonio and Nazas poles have been rotated about pole of rotation suggested by tectonic model of Anderson and Schmidt (1983).

model of Anderson and Schmidt (1983) calls for an angle of $-15°$, which is based on both geological and geometrical constraints.

Both models show an improvement in the clustering of poles for each unit compared to the undisplaced poles. If McElhinny's test (1964) is applied to determine whether or not this improvement is statistically meaningful, it is significant at the 95% level for Anderson and Schmidt's (1983) model ($k_2/k_1 = 5.1$ with $N = 4$) but is not significant for Scotese et al.'s (1979) model ($k_2/k_1 = 1.34$; see Fig. 10).

The four poles in Fig. 10, as adjusted for Anderson and Schmidt's (1983) model, may be compared to cratonic North American paleopoles for 200 and 190 Ma (Irving, 1979). The displacement of the restored poles from Irving's (1979) Triassic–Jurassic poles is slight and may be represented as a pure rotation of approximately 8° clockwise about the sampling sites (see Fig. 11). This could arise from an undetected systematic error in our data, but it seems odd that Nairn's (1976) data should agree so well with ours if this is the case. A minor bending or rotation of our study area of a few degrees with respect to cratonic North America

is a possibility that might bear further investigation. A third alternative arises from Irving's (1979) use of a smoothed pole over an interval of 10 m.y. at a cusp in the apparent polar wander path. This could displace his average pole to the east. There is a number of individual Triassic–Jurassic poles, however, which agree quite well with Irving's (1979) average pole, so this effect might not be important.

At this time we have no particular preference for any one of these alternatives.

CONCLUSIONS

If Triassic–Jurassic paleomagnetic poles from the Luning and Sil Nakya formations north of the proposed Mojave–Sonora megashear are compared with poles from the equivalent Antimonio and Nazas formations south of the megashear, all four poles are statistically indistinguishable from one another at the 95% confidence level both before and after proposed displacement along the megashear is restored. Nonetheless, the improvement in grouping of the poles after restoration of the displacement according to the model of Anderson and Schmidt (1983) is sufficient to indicate at the 95% confidence level that the two sets of poles are not in fact drawn from the same distribution of poles. The apparent contradiction between the two statistical conclusions may well arise from our very conservative application of Fisher statistics to the formation poles. As an example, the α_{95} for the Antimonio Formation is based on use of $N = 3$ (number of localities) rather than on $N = 19$ (number of sites) or $N = 115$ (number of specimens). Substantially smaller values of α_{95} would have resulted from either of the latter choices.

The comparisons of the four poles considered in this study with Irving's (1979) 190 Ma poles reveal a distinct displacement of the former from the latter. Systematic errors, a slight clockwise rotation of northern Mexico and the southern U.S. Cordillera with respect to cratonic North America, or the inappropriateness of Irving's (1979) smoothed poles near the Triassic crusp in the apparent polar wander curve are all possible explanations.

The Cretaceous results of this study are based on fewer samples and provide less of a constraint on the motion along the megashear. Nonetheless, the poles from the Sand Wells Formation (north of the megashear) and the unnamed volcanics of locality 7 (south of the megashear) are in good agreement with one another, suggesting that motion along the megashear had ceased by the Cretaceous. The two poles are displaced somewhat from the North American Cretaceous mean poles, and to a greater extent from the poles for the Roskruge volcanics in Arizona and the Difunta Group in Mexico. These displacements appear to be essentially random and do not shed light on any existing tectonic model. They may result from the small number of sites used in our Cretaceous results. Additional collection is clearly called for.

These results might properly be interpreted as a strong suggestion that the postulated displacement along the megashear did occur, but not as a convincing

proof. There is also an indication that displacement had ceased by the Cretaceous, as postulated on geochronological and geological grounds by Anderson and Silver (1979).

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